16 March 2026

Exercise 3: SQL CASE Statements

1. Classify each product by price:
   - Expensive if price >1000
   - Mid-range if price between 100 and 1000
   - Budget if price < 100

Expected output columns: product_name, price, price_category

```
SELECT product_name,
       price,
CASE
WHEN price > 1000 THEN 'Expensive'
WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'
ELSE 'Budget'
END AS price_category
FROM products;
```

product_name	price	price_category
Laptop	1200.00	Expensive
Phone	800.00	Mid-range
Monitor	300.00	Mid-range
Keyboard	45.00	Budget
Mouse	25.00	Budget

2. Label each order:
   - High value for orders ≥ 1000
   - Medium Value for 500 - 999.99
   - Low value for orders < 500

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```
SELECT customer_name,
       amount,
CASE
 WHEN amount >= 1000 THEN 'High value'
 WHEN amount BETWEEN 500 AND 999.99 THEN 'Medium value'
 ELSE 'Low value'
 END AS order_value_category
 FROM orders;
```

Customer_name	amount	Order_value_category
Alice	150.00	Low value
Bob	560.00	Medium value
Charlie	999.99	Medium value
Diana	45.50	Low value
Ethan	1200.00	High value

3. Categorize employee position
   - If in IT and salary > 80000 → 'Senior IT'
   - If in HR and salary > 55000 → 'Experienced HR'
   - Otherwise → 'Staff'

Expected columns: emp_name, department, salary, position_level

```
SELECT emp_name,
       department,
       salary,
CASE
WHEN department = 'IT' AND salary > 80000 THEN 'Senior IT'
WHEN department = 'HR' AND salary > 55000 THEN 'Experend HR'
ELSE 'Staff'
END AS position_level
FROM employees;
```

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emp_name	department	Salary	position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

4. Assign a letter grade ≥90 : A , 80-89:B 70-79:C
60-69:D <60:F.

Expected output column: student_name ,score, grade

```
SELECT student_name,
       score,
CASE
WHEN score >=90 THEN 'A'
WHEN score BETWEEN 80 AND 89 THEN 'B'
WHEN score BETWEEN 70 AND 79 THEN 'C'
ELSE 'D'
END AS grade
FROM students;
```

student_name	score	grade
Anna	92	A
Ben	76	C
Cara	59	D
David	83	B
Ella	68	D

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5. Label delivery performance:

- ≤ 30 mins: Fast
- 31 - 60 mins : On time
- 60 mins : Late

Expected output columns: delivery_id, delivery_time_minutes, performance

```
SELECT delivery_id,
       delivery_time_minutes
CASE
WHEN delivery_time_minute <= 30 THEN 'Fast'
WHEN delivery_time_minute BETWEEN 31 AND 60 THEN 'On time'
ELSE Late
END AS performance
FROM deliveries;
```

delivery_id	delivery_time_minutes	
1	45	On time
2	80	Late
3	30	Fast
4	65	Late
5	100	Laste

6. Convert priority to labels

- 3 -> High
- 2 -> Medium
- 1 -> Low

Expected output columns: issue_type, priority, priority_label

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```
SELECT issue_type,
       priority,
CASE
WHEN priority = 3 THEN 'High'
WHEN priority = 2 THEN 'Medium'
ELSE 'Low'
END AS priority_label
FROM tickets;
```

issue_type	priority	priority_label
Login issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

7. Calculate attendance % and classify:
- ≥ 90% → Excellent
- 75 - 89% → Good
- < 75% → Needs Improvement

Expected output columns: student_id, attendance_percentage, attendance_status

```
SELECT student_id,
       SUM(days_present/total_days) AS attendance_percentage,
       CASE
       WHEN SUM(days_present/total_days) >= 0.9 THEN 'Exellent'
       WHEN SUM(days_present/total_days) BETWEEN 0.75 AND 0.89
       THEN 'Good'
       ELSE 'Need Improvement'
       END AS attendance_status
       FROM attendance
       GROUP BY student_id;
```

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Student_id	attendance_percentage	attendance_status
1	0.9	Excellent
2	0.6	Need Improvement
3	0.96	Excellent
4	0.5	Need Improvement
5	1	Excellent

8. Label stock status:

- 0 → Out of stock
- 1-5 → Low stock
- 5 → In stock

Expected output columns: product_id, stock_qty, stock_status

```
SELECT product_id,
       stock_qty
CASE
WHEN stock_qty = 0 THEN 'Out of Stock'
WHEN stock_qty BETWEEN 1 AND 5 THEN 'Low Stock'
ELSE 'In stock'
END AS stock_status
FROM products_inventory;
```

product_id	stock_qty	stock_status
1	5	Low Stock
2	0	Out of stock
3	25	In stock
4	10	In stock
5	3	Low Stock

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9. Classify by size :

- $\geq 25 \rightarrow$ Large
- $10 - 24 \rightarrow$ Medium
- $< 10 \rightarrow$ Small

Expected output columns: Subject, enrolled_students, class_size_category

```
SELECT subject,
       enrolled_students,
CASE
WHEN enrolled_students >= 25 THEN 'Large'
WHEN enrolled_students BETWEEN 10 AND 24 THEN 'Medium'
ELSE 'Small'
END AS class_size_category
FROM classes;
```

Subject	enrolled_students	class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

10. Apply discount flag:

- If payment_method = Cash and amount $\geq 200 \rightarrow$ Eligible for Discount
- Otherwise $\rightarrow$ Not Eligible

Expected Output columns: payment_id, payment_method, amount, discount_eligibility

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```
SELECT payment_id,
       payment_method,
       amount,
CASE
WHEN payment_method = 'Cash' AND amount >= 200 THEN
'Eligible for Discount'
ELSE 'Not Eligible
END AS discount_eligibility
FROM payments;
```

payment_id	payment_method	amount	discount_eligibility
1	Card	50	Not Eligible
2	Cash	200	Eligible for Discount
3	Card	150	Not Eligible
4	PayPal	75	Not Eligible
5	Cash	300	Eligible for Discount

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